

# Psychometric Evaluation of the Mental Health Continuum–Short Form in French Canadian Young Adults

Évaluation psychométrique du Continuum de santé mentale–version abrégée (CSM-VA) chez des jeunes adultes canadiens-français

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The Canadian Journal of Psychiatry /  
La Revue Canadienne de Psychiatrie  
2017, Vol. 62(4) 286-294  
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DOI: 10.1177/0706743716675855  
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## Abstract

**Objective:** To examine the factor structure, internal consistency, reliability, sex invariance, and discriminant validity of the French Canadian version of the Mental Health Continuum–Short Form (MHC-SF).

**Method:** A total of 1485 French-speaking postsecondary students in Quebec, Canada (58% female; mean age = 18.4, SD = 2.4), completed the MHC-SF. Confirmatory factor analysis (CFA) was used to assess the factorial structure of the MHC-SF. Internal consistency was assessed with Cronbach’s alpha, and reliability was assessed with the rho reliability coefficient. Invariance testing across sex was conducted using multigroup CFA comparing 4 increasingly restrictive models, and discriminant validity was examined against the Hospital Anxiety and Depression Scale (HADS) using Pearson correlation coefficients and CFA.

**Results:** CFA supported the correlated 3-factor structure of the MHC-SF, with emotional, social, and psychological well-being subscales. The scale and each subscale items had internal consistency coefficients (Cronbach’s alphas) above .70 and reliability coefficients (Jöreskog’s rho) ranging from .79 to .90. Based on the multigroup CFA, configural, metric, scalar, and error variance invariance of the MHC-SF was observed across sex. Finally, the 2-continua model, suggesting that mental health and mental illness are distinct but related dimensions, was supported by both moderate inverse correlations between MHC-SF and HADS subscale scores and the 2-factor structure in CFA.

**Conclusions:** These data support the multidimensional structure of the MHC-SF and provide evidence of internal consistency, reliability, and invariance across sex. The MHC-SF is a valid and reliable measure of mental health that is distinct from mental illness among French Canadian young adults.

## Abrégé

**Objectif :** Examiner la structure factorielle, la cohérence interne, la fiabilité, l’invariance entre les sexes, et la validité discriminante de la version canadienne-française du Continuum de santé mentale-version abrégée (CSM-VA) (Mental Health Continuum–Short Form).

**Méthode :** Un total de 1 485 étudiants francophones aux études post-secondaires du Québec, au Canada (58% femmes; âge moyen = 18,4, ET = 2,4) ont répondu au CSM-VA. L’analyse factorielle confirmatoire (AFC) a servi à évaluer la structure

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factorielle du CSM-VA. La cohérence interne a été estimée à l'aide de l'alpha de Cronbach, et la fiabilité a été évaluée par le coefficient de fiabilité  $\rho$ . L'invariance selon le sexe a été évaluée à l'aide de l'AFC multi-groupe comparant quatre modèles de plus en plus restrictifs, et la validité discriminante a été examinée par rapport à la Hospital Anxiety and Depression Scale (HADS) à l'aide des coefficients de corrélation de Pearson et de l'AFC.

**Résultats :** L'AFC confirme la structure à trois facteurs corrélée du CSM-VA qui correspondent aux sous-échelles de bien-être émotionnel, psychologique et sociale. L'échelle globale et les sous-échelles ont des coefficients de consistance interne (alphas de Cronbach) supérieurs à 0,70 et des coefficients de fiabilité ( $\rho$  de Jöreskog) qui varient de 0,79 à 0,90. D'après l'AFC multi-groupe, l'invariance configurale, métrique, scalaire et de variance-erreur du CSM-VA a été observée entre les sexes. Enfin le modèle des deux continuums, suggérant que la santé mentale et la maladie mentale sont des dimensions distinctes mais reliées est confirmée par des corrélations inverses modérées entre les scores du CSM-VA et les sous-échelles du HADS et par la structure bifactorielle identifiée dans l'AFC.

**Conclusions :** Ces données confirment la structure multidimensionnelle du CSM-VA, soutiennent la cohérence interne, la fiabilité et l'invariance entre les sexes de l'échelle. Le CSM-VA est une mesure valide et fiable de la santé mentale, distincte de l'évaluation de la maladie mentale chez les jeunes adultes canadiens-français.

### Keywords

Mental Health Continuum, psychometric, French Canadian version, common mental disorders, measurement invariance, youth

Mental health has traditionally been conceptualised as the absence of mental illness.<sup>1,2</sup> However, recently, a more holistic characterisation suggests that mental health is “a state of well-being in which every individual realises his/her potential, can cope with the normal stresses of life, can work productively and fruitfully, and is able to make a contribution to her or his community.”<sup>3</sup> Thus, mental health is not the absence of mental illness but rather the foundation of well-being and effective functioning at an individual and social level.<sup>1</sup> Interest in mental health as distinct from mental illness is increasing among researchers, clinicians, and public health practitioners. To fully capture the state of mental health, its assessment should integrate the spectrum of well-being and positive functioning, in addition to mental illness. The Mental Health Continuum–Short Form (MHC-SF)<sup>4</sup> provides such a comprehensive assessment by measuring emotional, psychological, and social well-being.

*Emotional well-being*, the first MHC-SF component, refers to the hedonic approach whereby well-being reflects affective reactions expressed by moods and emotions. Affect describes the extent to which people experience a sense of wellness in their lives. Increasing mental well-being implies maximising positive while minimising negative affect.<sup>5</sup> Life satisfaction, a component of hedonic well-being, refers to the cognitive global judgment of events occurring in one's life.<sup>6</sup>

*Psychological and social well-being*, two additional MHC-SF components, refer to the eudaimonic approach,<sup>6</sup> wherein well-being comprises the capacity of actualising human potential, resulting in positive functioning at individual and social levels. Assessment of individual functioning is based on Ryff's multidimensional model,<sup>7</sup> which operationalises psychological well-being as personal challenges encountered as individuals strive to realise their potential. This model incorporates 6 dimensions of psychological well-

being: self-acceptance, positive relations with others, personal growth, purpose in life, environmental mastery, and autonomy. Assessment of social functioning is based on Keyes's multidimensional model<sup>8</sup> to evaluate the social challenges and tasks of individual functioning that examines interpersonal relations and adjustments in social life. This model comprises 5 dimensions: social coherence, social actualisation, social integration, social acceptance, and social contribution.

Together, the hedonic and eudaimonic perspectives capture the positive spectrum of mental health.<sup>1,4</sup> Although their measures and constructs overlap, theoretical and empirical research has demonstrated that they are not redundant.<sup>9</sup> However, the structure of well-being has been debated recently,<sup>10,11</sup> and empirical research suggests that these are distinct approaches to well-being but not distinct constructs; thus, they are better represented by single ‘general’ factor of well-being.<sup>12-14</sup>

Accordingly, mental health and illness are not extremes of one continuum but distinct (although correlated) continua.<sup>4</sup> The two-continua model posits one continuum indicating level of mental health, while the other refers to the presence or absence of mental illness. Optimal mental health is characterised by the absence of mental illness and a high level of mental health.<sup>1</sup> Individuals may, at the same time, have mental illness yet experience high levels of mental health. Conversely, individuals without mental illness may have low mental health and experience poor psychosocial functioning, high health care utilisation, and low work productivity.<sup>15</sup> An assessment of both mental health and illness has predicted psychosocial functioning better than a mental illness diagnosis alone, supporting their complementarity in providing accurate assessments of mental state.<sup>4</sup>

The MHC-SF is highly attractive for use in research, surveillance, and clinical settings. It has been translated into

several languages and validated across cultural contexts in North America, Europe, Africa, and Asia.<sup>4,16-27</sup> Studies using exploratory structural equation modeling (ESEM)<sup>20,21</sup> and exploratory and/or confirmatory factor analyses support its 3-factor structure in adolescents<sup>24-26</sup> and adults.<sup>16,18,19,22,28</sup> A second-order structure, with a general mental health latent factor, fit the data in a manner equivalent to that of the first-order 3-factor model.<sup>17</sup> A bifactor structure, with 1 general factor (i.e., well-being) and 3 specific factors, provided the best fit in a Serbian sample,<sup>27</sup> while a Brazilian study indicated that a single dimension is sufficient to represent the structure.<sup>23</sup>

Sex invariance of the MHC-SF (i.e., whether scores have equivalent meaning across sex<sup>29</sup>) has been studied<sup>16,17</sup> but not specifically among youth in transition to adulthood. Sex is a determinant of mental state. It relates to differential susceptibility; exposure to risk<sup>30</sup> and prevalence of depression and anxiety, especially among youth<sup>31</sup>; and disparities in subjective well-being.<sup>32</sup> Because sex differences in mental health are of major interest for clinicians, researchers, and public health surveillance and intervention planning, sex invariance of the MHC-SF should be tested. If scores do not exhibit invariance, comparison across sex may lead to biased interpretations of results.<sup>33</sup>

Finally, numerous studies comparing MHC-SF scores with mental disorder symptoms<sup>16-19,23,24,34</sup> confirm the plausibility of the 2-continua model.

This article examines the psychometric properties of the French Canadian MHC-SF in young adults through 4 objectives: 1) to test the factor structure of the MHC-SF, 2) to assess the internal consistency and reliability of MHC-SF subscales, 3) to examine sex invariance of the MHC-SF, and 4) to assess its discriminant validity against the Hospital Anxiety and Depression Scale (HADS). We hypothesise that 1) the MHC-SF is multidimensional, with 3 factors representing emotional, social, and psychological well-being as observed in numerous studies<sup>16,18,19,20-22,24-26</sup>; 2) the 3 factors tap a general well-being factor and thus a higher order latent factor will be empirically valid, as suggested in an Italian study<sup>17</sup>; 3) MHC-SF subscales have high reliability and are sex invariant, similar to earlier findings<sup>16,17,21,22,26</sup>; and 4) mental health and illness represent distinct yet correlated factors as found previously.<sup>16-19,23,24,34</sup>

The study population includes youth in transition to adulthood, an intense developmental period with new challenges related to social role changes, including negotiating transitions in education, employment, independent housing, family formation, and parenthood.<sup>34</sup> Because this group is targeted for mental health promotion strategies, identification of reliable and valid measures of mental health is a priority.

## Method

Participants included 1485 postsecondary students (58% female; age 16 to 40 y [ $M = 18.4$ ,  $SD = 2.4$ ], 97.4% were

aged 16 to 24 y) recruited in a CEGEP (Quebec, Canada). In Quebec, CEGEPs provide postsecondary school education, including 2-y preuniversity programs and 3-y career programs, which typically lead to employment. Data were collected in 88 of 103 (85.4%) compulsory physical education classes during October 2013. Data collection was not possible in 13 weekend classes, and scheduling precluded data collection in another 2 classes. Those present during data collection were invited by their teacher to participate; 1527 of 1746 eligible students (87.4%) provided informed consent and completed paper-and-pencil questionnaires during class time. All documents were returned to the teacher in sealed envelopes to preserve anonymity. Participants did not receive compensation. A subset of 1485 participants who completed all 14 MHC-SF items (97.2%) constituted the analytic sample. There were no significant differences in age or sex between students with ( $n = 42$ , 2.8%) and without ( $n = 1485$ ) missing data. Discriminant validity was examined among participants with complete data on the MHC-SF, HADS, and sex ( $n = 1457$ , 95.4%); there were no significant differences in age or sex between included and excluded participants ( $n = 70$ , 4.6%).

## Measures

The MHC-SF was translated into French with back translation into English to ensure equivalency of each item (Supplemental Appendix I).<sup>35</sup> The 14-item MHC-SF includes 3 items measuring emotional well-being (items 1 to 3) defined in terms of positive affect and satisfaction in life, 5 items measuring social well-being (items 4 to 8) according to the dimensions described in Keyes's social well-being model, and 6 items measuring psychological well-being (items 9 to 14), 1 item for each dimension identified in Ryff's model. Participants rated how often they felt a certain way during the last month on a 6-point Likert scale (0 to 5): *never*, *rarely*, *a few times*, *often*, *most of the time*, and *all the time*. Scores can be computed for the overall scale (range, 0 to 70) and for each subscale (i.e., emotional [range, 0 to 15], social [range, 0 to 25], and psychological [range, 0 to 30] well-being).

The HADS, a widely used brief questionnaire that identifies possible and probable cases of anxiety and depressive disorders, has good sensitivity and specificity in both primary care patients and the general population.<sup>36</sup> It comprises 14 items using the previous 7 days as a reference period and includes an Anxiety subscale (HADS-A) and a Depression subscale (HADS-D), both containing 7 items. Each item is scored on a 4-point Likert scale indicating absence (0), possible presence (1 to 2), or presence of anxiety or depressive symptoms (3). The total score ranges between 0 and 42 (0 to 21 for each subscale). The French Canadian version has shown internal consistency coefficients (Cronbach's alphas = .82 to .89). A 2-factor structure has been identified, reflecting anxiety and depression factors in a large French Canadian primary care sample.<sup>37</sup>

## Data Analysis

Confirmatory factor analysis (CFA) was performed to assess the internal structure of the MHC-SF. Based on theory and previous research using different MHC-SF versions,<sup>1,4</sup> 4 models were compared: 1) a single-factor structure, in which a single dimension describing general well-being is identified; 2) a 2-correlated factor structure, wherein one factor represents the hedonic dimension and the other factor represents the eudaimonic dimension; 3) a 3-correlated factor structure, wherein the factors represent the emotional, social, and psychological well-being; and 4) a second-order model with a general well-being latent factor encompassing the 3 first-order factors. As reported,<sup>1,4,18</sup> we expected the factors to be correlated. We examined the pattern matrix of item loadings; .40 was used as the cut-point for acceptable factor loadings.

The parameter estimates in CFA were obtained using the robust maximum likelihood method with the Satorra-Bentler scaled chi-square ( $SB \chi^2$ ) because the assumption of multivariate normality was not fulfilled; Mardia's coefficient of multivariate skewness and kurtosis were 35.79 and 23.56, respectively ( $P = 0.000$ ). As recommended by Chen,<sup>38</sup> several fit indices were computed to assess whether the theoretical and empirical models were consistent with the data, in addition to the  $SB \chi^2$ : the root mean square error of approximation (RMSEA), the comparative fit index (CFI), the standardised root mean square residual (SRMR), and the Tucker-Lewis index (TLI). Values  $<.08$  for RMSEA and SRMR,  $>.95$  for CFI, and  $>.90$  for TLI were used to make decisions about model fit.<sup>38</sup>

The internal consistency of MHC-SF subscales was examined using Cronbach's alpha coefficient. Given that the alpha coefficient is based on strict assumptions (e.g., unidimensionality, uncorrelated errors, and essential tau-equivalence of all items [all factor loadings and all error variances are constrained to be equal]) that are often violated, it may over- or underestimate reliability.<sup>39</sup> Therefore, a composite reliability coefficient (Jöreskog's rho) was used to test if a single common factor underlies the MHC-SF. The convergent validity rho (average variance extract [AVE]) was also computed to measure the amount of variance captured by the MHC-SF in relation to the variance due to random measurement error.<sup>40</sup> Jöreskog's rho values above .70 and AVE  $>.50$  were used as cut-points for assessing acceptable coefficients.<sup>40</sup>

Sex invariance of the best-fitting MHC-SF model was examined in multigroup confirmatory factor analysis (MG-CFA), which tested 4 levels of measurement invariance by comparing increasingly restrictive models.<sup>29</sup> Each new model is nested in the previous model. This strategy is recognised as the most powerful and versatile approach for testing invariance.<sup>41</sup> Specifically, *configural invariance* (model 1) constrains the model structure as equal, which implies that the number of factors and pattern of factor-item loadings are the same across sex. Configural invariance is a prerequisite for further invariance testing. *Metric invariance* (model 2) adds constraints on factor loadings; the indicators should relate to

the factors in the same way across sex, which provides evidence of equal-pattern coefficients. This stronger level of factorial invariance tests whether participants across sex attribute the same meaning to the latent constructs studied. *Scalar invariance* (model 3) constrains intercepts of the items to be equal across sex. Scalar invariance suggests that participants with equal scores on the latent construct obtain the same score on the observed variable, regardless of sex. If scalar invariance is satisfied, MHC-SF scores can be compared across sex, and observed item differences will indicate sex differences on the latent construct.<sup>42,43</sup> *Error variance invariance* (model 4), the last and most restrictive model, constrains all error variances to be equal across sex and tests if each item has the same level of measurement error between groups.

Robust maximum likelihood estimation was used in MG-CFA analyses. For each increasingly restrictive invariance model tested, several goodness-of-fit indices are reported:  $SB \chi^2$  ( $df$ ), CFI, RMSEA, SRMR, and TLI. Configural invariance (model 1) is claimed if RMSEA and SRMR values  $<.08$  and TLI  $>.90$ , supplemented by CFI values  $>.95$ .<sup>33</sup> If configural invariance is attained, comparisons with more restricted models are performed. As recommended by Chen,<sup>38</sup> metric, scalar, and error variance invariance is examined as changes in CFI ( $\Delta CFI$ ) and RMSEA ( $\Delta RMSEA$ ) between nested models. Absolute differences  $\leq .01$  in CFI and  $\leq .015$  in RMSEA indicate model invariance.<sup>33</sup>

Pearson correlation coefficients between MHC-SF and the HADS subscales were examined to assess the plausibility of the 2-continua model. Both the magnitude and direction of the coefficients were examined; a  $P$  value  $<.05$  denoted a statistically significant relationship between variables, and a negative coefficient indicated an inverse relationship. Correlations were computed for the total sample and for males and females separately. Inverse high correlation close to  $-1$  supported the hypothesis of mental health and mental illness as extremes of a single continuum; inverse moderate correlations suggested that these constructs represent a different continuum, supporting the 2-continua model. In addition to the Pearson correlations, CFA was performed to identify the most appropriate model. The single-factor model assumed that all subscales represent a single, bipolar latent dimension (confirming that the absence of mental illness implies the presence of mental health). This model was compared to a 2-factor model, which allows factors to be correlated, positing that mental health and mental illness subscales represent 2 latent constructs. Several fit indices were compared. Analyses were undertaken using SPSS 20 (SPSS, Inc., an IBM Company, Chicago, IL), Lisrel (Scientific Software International, Inc., Skokie, IL), and R Studio (RStudio, Boston, MA).

## Results

### Factor Structure

The fit indices for each model are presented in Table 1. The intercorrelated 3-factor model provided the best fit to the

**Table 1.** Fit Indices from the Confirmatory Factor Analysis for 4 Competing Models of Mental Well-Being ( $n = 1485$ ).

| Models                       | SB $\chi^2$ | df | CFI | RMSEA | SRMR | TLI |
|------------------------------|-------------|----|-----|-------|------|-----|
| One factor                   | 1411.20     | 77 | .84 | .12   | .07  | .81 |
| Two factors                  | 895.80      | 76 | .90 | .10   | .06  | .88 |
| Three factors                | 527.96      | 74 | .94 | .07   | .05  | .93 |
| Second order (three factors) | 527.96      | 70 | .94 | .07   | .05  | .93 |

CFI, confirmatory fit index; df, degrees of freedom; RMSEA, root mean square error of approximation; SB  $\chi^2$ , Satorra-Bentler scaled chi-square; SRMR, standardised root mean squared of the residuals; TLI, Tucker-Lewis index.

**Table 2.** Factor Loadings of the 3–Correlated Factor Model of the MHC-SF ( $n = 1485$ ).

| MHC-SF Subscales and Item Number and Content | Standardised Regression Weights for Each Factor |     |     |
|----------------------------------------------|-------------------------------------------------|-----|-----|
|                                              | I                                               | II  | III |
| I. Emotional well-being                      |                                                 |     |     |
| 1. Happiness                                 |                                                 |     | .82 |
| 2. Interest in life                          |                                                 |     | .89 |
| 3. Life satisfaction                         |                                                 |     | .89 |
| II. Social well-being                        |                                                 |     |     |
| 4. Social contribution                       |                                                 | .64 |     |
| 5. Social integration                        |                                                 | .55 |     |
| 6. Social actualisation                      |                                                 | .80 |     |
| 7. Social acceptance                         |                                                 | .63 |     |
| 8. Social coherence                          |                                                 | .65 |     |
| III. Psychological well-being                |                                                 |     |     |
| 9. Self-acceptance                           | .75                                             |     |     |
| 10. Environment mastery                      | .68                                             |     |     |
| 11. Positive relations with others           | .67                                             |     |     |
| 12. Personal growth                          | .70                                             |     |     |
| 13. Autonomy                                 | .59                                             |     |     |
| 14. Purpose in life                          | .77                                             |     |     |

MHC-SF, Mental Health Continuum–Short Form.

data. In the best-fitting model, all items loaded significantly on their expected factors (Table 2). The higher order model fit the data as well as the first-order 3-factor model, providing support for a general well-being factor encompassing the 3 first-order factors representing emotional, social, and psychological well-being (Supplemental Appendix II presents path diagrams). The correlation was .82 between the latent factors representing emotional and psychological well-being in CFA, .74 between social and psychological well-being factors, and .66 between emotional and social well-being factors. Computation of the determination coefficient ( $R^2$ ) revealed that 70% of the variance was shared by the most highly correlated factors (emotional and psychological well-being), 55% by social and psychological factors, and 44% by emotional and social well-being factors, suggesting that these factors capture different dimensions of well-being.

## Internal Consistency and Reliability

The 3 MHC-SF subscales (emotional, psychological, social) had internal consistency Cronbach's alpha coefficients ranging from .78 to .90 (Table 3). The composite reliability coefficients (Jöreskog's rho) were also above .70, whereas the AVE (or rho vc) ranged from .48 to .75. All 3 subscales were highly correlated with the overall MHC-SF score and were also highly intercorrelated (Table 3).

## Measurement Invariance

The 3-factor model fit the data well across sexes: male, SB  $\chi^2(df) = 246.96(74)$ , CFI = 0.94, RMSEA = 0.07, SRMR = 0.05, TLI = 0.93; female, SB  $\chi^2(df) = 344.11(74)$ , CFI = 0.95, RMSEA = 0.07, SRMR = 0.05, TLI = 0.94. The analyses for configural invariance showed an acceptable baseline model according to the CFI, RMSEA, SRMR, and TLI (Table 4). Based on  $\Delta$ CFI and  $\Delta$ RMSEA values below the thresholds of .010 and .015, respectively, metric, scalar, and error variance invariance of the model was supported across sexes.

## Discriminant Validity: The 2-Continua Model

Table 5 shows correlations between MHC-SF and HADS subscales for the total sample and by sex. HADS anxiety and depression subscales were negatively correlated with all 3 MHC-SF subscales as well as with the total scale in the total sample and in both males and females. Higher levels of anxiety and depressive symptoms were linked to lower levels of emotional, social, and psychological well-being. Correlations ranged from  $-.24$  for social well-being and anxiety symptoms in males to  $-.56$  for emotional well-being and depressive symptoms in females. The CFA also supported the model with 2 correlated factors ( $m_2$ ) as indicated by better fit coefficients compared to the single-factor ( $m_1$ ) model:  $m_2$ , CFI = .99, RMSEA = .10, SRMR = .02, TLI = .95;  $m_1$ , CFI = .94, RMSEA = .16, SRMR = .05, TLI = .88. The 2 latent factors were inversely correlated ( $r = -.78$ ), and all items loaded significantly on their expected factors (Supplemental Appendix III). These findings support the hypothesis that mental health and illness are distinct yet related constructs.

## Discussion

The present study examined the factor structure of MHC-SF French Canadian version, its internal consistency and reliability, its invariance across sex, and the plausibility of the 2-continua model proposing that mental health and illness are distinct yet related constructs.

Supporting previous studies of the MHC-SF across language and cultures,<sup>16,18-22,24-26,28</sup> the CFA revealed that the 3-factor model displayed the best fit to the data. The 3-factor structure corresponds to the 3 core components of the World Health Organization (WHO) definition of mental health:

**Table 3.** Descriptive Statistics for the Overall MHC-SF Scale and Each of Its 3 Subscales ( $n = 1485$ ).

| MHC-SF Scale and Subscales  | Cronbach's Alpha | Jöreskog's Rho | Rho vc | Pearson Correlation Coefficients |     |     |     |
|-----------------------------|------------------|----------------|--------|----------------------------------|-----|-----|-----|
|                             |                  |                |        | 1                                | 2   | 3   | 4   |
| 1. Overall MHC-SF           | —                | —              | —      | 1                                | .83 | .86 | .91 |
| 2. Emotional well-being     | .90              | .90            | .75    |                                  | 1   | .57 | .71 |
| 3. Social well-being        | .78              | .79            | .43    |                                  |     | 1   | .61 |
| 4. Psychological well-being | .85              | .85            | .48    |                                  |     |     | 1   |

MHC-SF, Mental Health Continuum–Short form.

**Table 4.** Fit Indices for Analyses Testing the Sex Invariance of the Intercorrelated 3-Factor Model ( $n = 1485$ ).

| Invariance Models                  | SB $\chi^2$ | df  | CFI  | $\Delta$ CFI | RMSEA | $\Delta$ RMSEA | SRMR | TLI  |
|------------------------------------|-------------|-----|------|--------------|-------|----------------|------|------|
| Model 1: Configural invariance     | 589.03      | 148 | .946 | —            | .071  | —              | .047 | .933 |
| Model 2: Metric invariance         | 604.71      | 161 | .946 | .000         | .068  | .003           | .048 | .939 |
| Model 3: Scalar invariance         | 690.55      | 171 | .938 | .008         | .071  | .003           | .050 | .934 |
| Model 4: Error variance invariance | 704.22      | 185 | .936 | .002         | .069  | .002           | .052 | .937 |

CFI, confirmatory fit index; df, degrees of freedom; RMSEA, root mean square error of approximation; SB  $\chi^2$ , Satorra-Bentler scaled chi-square; SRMR, standardised root mean squared of the residuals; TLI, Tucker-Lewis index.**Table 5.** Pearson Correlation Coefficients for the Associations between the MHC-SF and HADS Subscales ( $n = 1457$ ).

| MHC-SF Scale and Subscales  | HADS Subscales           |        |                  |        |                |        |
|-----------------------------|--------------------------|--------|------------------|--------|----------------|--------|
|                             | Total Sample, $n = 1457$ |        | Women, $n = 846$ |        | Men, $n = 611$ |        |
|                             | HADS-A                   | HADS-D | HADS-A           | HADS-D | HADS-A         | HADS-D |
| 1. Emotional well-being     | -.48                     | -.56   | -.49             | -.56   | -.41           | -.46   |
| 2. Social well-being        | -.33                     | -.38   | -.34             | -.41   | -.24           | -.32   |
| 3. Psychological well-being | -.40                     | -.51   | -.44             | -.51   | -.32           | -.47   |
| 4. Total MHC-SF             | -.45                     | -.54   | -.47             | -.55   | -.40           | -.53   |

All correlations were significant at  $P < 0.01$ .

HADS, Hospital Anxiety and Depressive Scale; HADS-A, Hospital Anxiety and Depressive Scale–Anxiety symptoms subscale; HADS-D, Hospital Anxiety and Depressive Scale–Depressive symptoms subscale; MHC-SF, Mental Health Continuum–Short Form.

emotional well-being and effective functioning at both individual and social levels.<sup>44</sup> As observed in at least one previous study,<sup>17</sup> our results suggest that the three factors derived from hedonic and eudaimonic dimensions are related to a more general construct of mental health as indicated by the second-order factor model. The emotional and psychological subscales of the French Canadian MHC-SF show high internal consistency and reliability, as assessed with Cronbach's alpha and Joreskog's rho. Internal consistency and reliability coefficients of the social well-being subscale, although satisfactory, were low relative to the other subscales. Similar findings have been observed in previous studies.<sup>16–19</sup> All Cronbach's alphas were above those observed in South African and Dutch and Italian studies.<sup>16,18,19</sup> As hypothesised, measurement invariance results provided evidence for the configural, metric, scalar, and error variance invariance of the French Canadian MHC-SF across sex. These findings suggest that MHC-SF is measured similarly across males and females using the 3-factor model, allowing for

comparison across sex. Analysis of the discriminant validity showed moderate to high inverse correlations between the MHC-SF and HADS subscales and also indicated that the 2–correlated factor model in CFA showed the best fit to the data. High correlations between subscales and highly correlated latent factors in CFA are not surprising since mental health and mental illness are known to be strongly related,<sup>4,44</sup> and mental health status is predictive of mental illness.<sup>45,46</sup> Although high, the correlations between subscales and latent factors do not approach 1, which would have suggested that these measure the same concept. Rather, these findings support the 2-continua model that mental health and mental illness are distinct but correlated constructs. This finding indicates that the absence of mental illness does not necessarily imply the presence of mental health, justifying the need for a measure to assess mental health.

Our study expands on previous research by examining the psychometric properties of the MHC-SF among youth in

transition to adulthood; other studies focus on adolescents<sup>24-26</sup> or adults.<sup>16,18-22,28</sup> Providing a valid mental health measure for this specific age group is critical for monitoring mental health to inform mental health research, policy, and care. The prevalence of common mental disorders and their potential long-term consequences on academic underachievement, substance abuse, and suicidal ideation, especially among youth,<sup>31,47</sup> justify the need to intervene to promote mental health and prevent mental disorders. By targeting youth transitioning to adulthood, we have an opportunity to intervene upstream, before the onset of mental disorders since most mental disorders appear in late adolescence or early adulthood.<sup>48</sup> Moreover, several studies confirm that changes in the level of mental health measured with the MHC-SF predict the risk of developing mental disorders.<sup>46,49-51</sup> Finally, the MHC-SF is sensitive to change, permitting evaluation of the effect of interventions to promote mental health.<sup>52</sup>

Limitations of this study include that the sample was restricted to postsecondary students in Quebec, which may limit generalisability of the findings. The cross-sectional data did not allow assessment of the temporal stability and sensitivity to change over time of the French Canadian MHC-SF; however, previous studies demonstrate that the instrument is stable over time but also sensitive to change, suggesting that MHC-SF scores are modifiable and can reflect and detect changes in mental health.<sup>17,19</sup> Also, even if the HADS is a valid, reliable, and widely used instrument to detect anxiety and depression symptoms, future research on the MHC-SF French Canadian version should test the 2-continua model with a structured psychiatric evaluation, which provides a complete assessment of mental disorders, as previously confirmed using the Composite International Diagnostic Interview-Short Form (CIDI-SF).<sup>4</sup>

## Conclusion

This study suggests that the French Canadian version of the MHC-SF is a valid and reliable brief self-report questionnaire to assess mental health. Future research should investigate whether the MHC-SF predicts academic performance, work productivity, school or workplace absenteeism, and health and illness outcomes.

## Acknowledgements

We thank the CEGEP de l'Outaouais.

## Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: ID is supported by doctoral awards from the 4P Program funded by the Canadian Institutes for Health Research and the Réseau de

recherche en santé des populations du Québec, the Fonds de recherche du Québec–Société et culture, the School of Public Health, and the Faculté des études supérieures et postdoctorales of the Université de Montréal. JOL holds a Canada Research Chair in the Early Determinants of Adult Chronic Disease. CMS holds a Canada Research Chair in Physical Activity and Mental Health.

## Ethical Approval

This study was approved by the CEGEP de l'Outaouais Research Ethics Board (approval number CER-2013-06-ID) and the University of Montreal Health Research Ethics Board (approval number 13-093-CERES-P). All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

## Supplemental Material

The online appendices are available at <http://journals.sagepub.com/doi/suppl/10.1177/0706743716675855>.

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